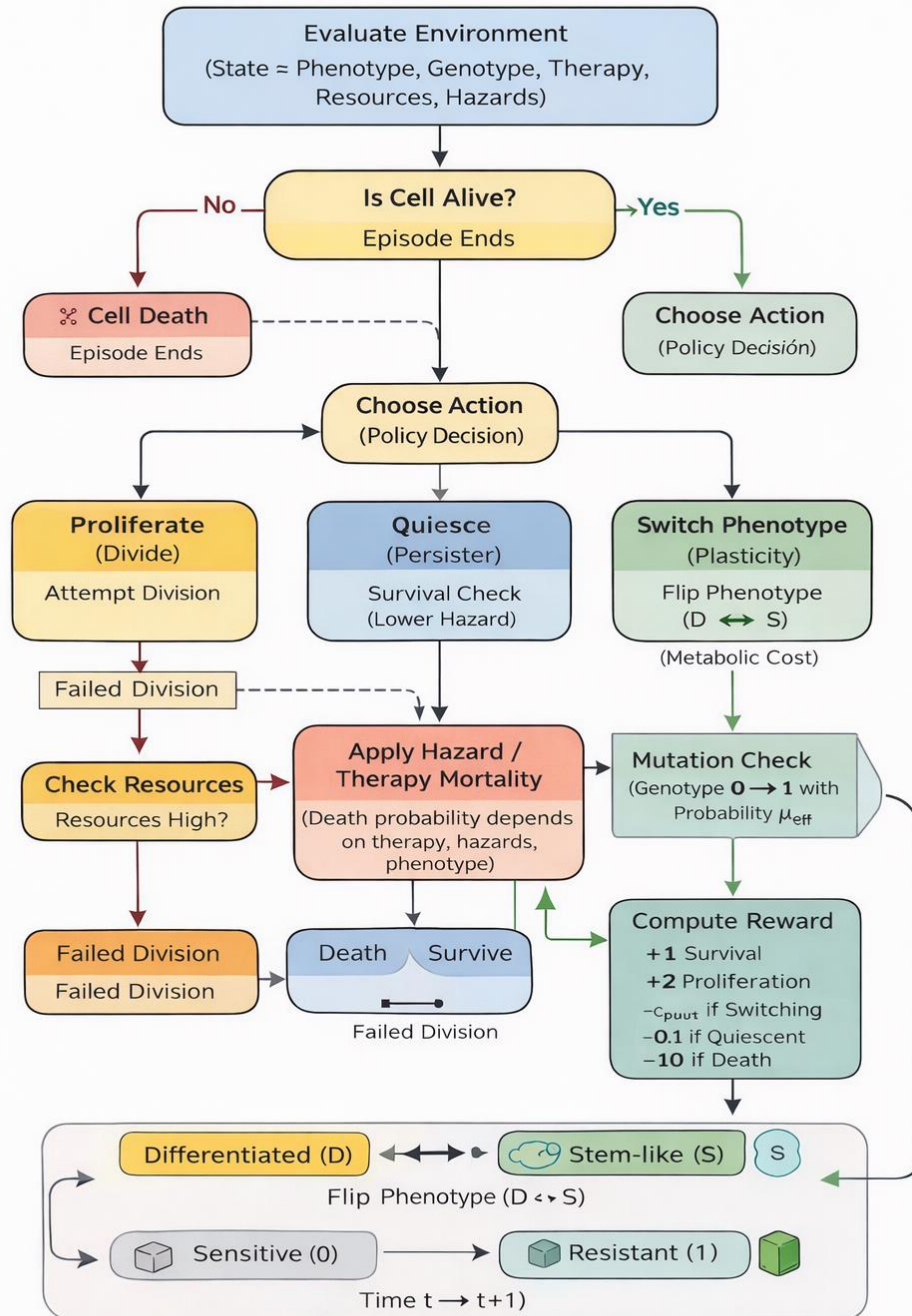


Cancer Cell Decision and Evolution Model



1 State space (what defines the MDP state)

A minimal but biologically meaningful state is:

State = (Phenotype, Genotype, Therapy, Resources, Hazards)

1. Phenotype (P):

- **S = stem-like / plastic / slow cycling / stress-tolerant**
- **D = differentiated / proliferative / therapy-sensitive**
Justification: stemness is treated as a functional state and can be modulated by ecology (e.g., hypoxia) rather than being fixed.

Classifying the evolutionary an...

2. Genotype (G):

- **0 = not genetically resistant**
- **1 = genetically resistant**
Justification: therapies select for resistance mutations; this is classic somatic evolution.

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3. Therapy (T) (adaptive therapy schedule):

- **ON / OFF** (or ON when tumor burden high, OFF when low)

4. Resources (R):

- **High / Low** (proxy for oxygen/nutrients/vasculature)
Justification: Eco-index resources include hypoxia, blood vessels, nutrients.

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5. Hazards (H):

- **High / Low** (proxy for therapy intensity + immune predation)
Justification: Eco-index hazards include immune infiltration and other threats.

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That gives $2 \times 2 \times 2 \times 2 \times 2 = 32$ states (small enough for Q-learning).

2 Action space (what the cell-agent can do)

At each time step (generation), the agent chooses one action:

1. Proliferate (A = PROLIF)

- increases tumor burden, increases chance of being hit by therapy hazards

2. Quiesce / Persister-like survival (A = QUIESC)

- no growth, lower hazard exposure

3. Plasticity switch (A = SWITCH)

- if $D \rightarrow S$ (dedifferentiate) or $S \rightarrow D$ (differentiate)

Justification: non-genetic heterogeneity can reflect **stable attractor states** in GRNs; switching = transition between attractors.

Reconciling Non-Genetic Plastic...

4. “Wait for mutation” is not a separate action

- it happens naturally during proliferation through mutation probability (see below).

3 Transition rules (biology dynamics)

A) Survival / death probability

Define baseline death probability as a function of hazards/resources and phenotype:

- If **Therapy ON** and **Hazards High**, then:
 - **D cells** have **high death probability**
 - **S cells** have **lower death probability** (stress tolerant)

This is consistent with the idea that harsh microenvironments and interventions impose strong selection and reduce diversity. For example, in the microenvironment-driven selection simulations, **low oxygen (harsh) reduces phenotype diversity and selects a few dominant phenotypes**.

Tumor Morphology and Phenotypic...

A simple rule table:

Condition	Death prob if D	Death prob if S
Therapy ON + Hazards High + Resources Low	very high	moderate
Therapy ON + Hazards High + Resources High	high	low-moderate
Therapy OFF + Hazards Low + Resources High	low	low
Therapy OFF + Resources Low (hypoxia)	moderate	low-moderate

(Your computational teammate can turn “low/moderate/high” into numbers like 0.05 / 0.15 / 0.35 / 0.6.)

B) Growth / reproduction (fitness benefit)

If the cell survives:

- **PROLIF** yields a positive growth reward **only if resources allow division**
- **S** typically divides slower than **D** (trade-off)

C) Genetic resistance emergence (Evo-index connection)

Resistance mutation occurs mainly when dividing:

- If action = PROLIF and genotype = 0:
 - genotype becomes 1 with probability μ_{eff}

Make μ_{eff} a function of Evo-index:

- **Higher D (diversity) and higher Δ (ongoing change)** → higher effective mutation supply / adaptation rate.
This matches the idea that Evo-index reflects evolvability as diversity + its change over time.

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(Implementation-friendly: set $\mu_{\text{eff}} = \mu_0 \times f(\text{Evo})$, where $f(\text{Evo})$ is 1 for low Evo and maybe 5–10× for high Evo.)

D) Plasticity switching (non-genetic evolution rule)

If action = SWITCH:

- phenotype flips ($D \leftrightarrow S$) **with high probability**
- BUT you pay a **plasticity cost** (this is the parameter you sweep to find tipping point)

Justification:

- Cell states can be stable attractors; switching is a transition between attractor states.

Reconciling Non-Genetic Plastic...

- Cancer stem-like states can reflect **non-genetic heterogeneity** and can be dynamic.

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4 Reward function (this is the key biology deliverable)

You want rewards that represent **fitness = survival + reproduction – costs** (Merlo/Maley definition of fitness as survival + reproduction is explicitly used in this literature).

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Use:

Reward = (Survival bonus) + (Proliferation bonus) – (Therapy penalty if killed) – (Plasticity metabolic cost) – (Quiescence opportunity cost)

Concrete version (easy to implement):

- Survive this step: **+1**
- If PROLIF and survive: **+b** (e.g., +0.5 to +2 depending on how strongly you reward growth)
- If die: **-10** (or end episode)
- If SWITCH: **-c_plast** ← **this is the parameter you sweep**
- If QUIESC: **-c_quiesc** (small penalty like 0.1 to reflect “no growth”)

Meaning of the tipping point:

As you increase **c_plast**, the learned optimal policy will stop using SWITCH and prefer PROLIF/QUIESC + eventually gain genetic resistance.